$f(t)$	$F(s)$
e^{at}	$\frac{1}{s-a}$
1	$\frac{1}{s}$
$\delta^{(n)}(t)$	s^n
t^n	$\frac{n!}{s^{n+1}}$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$e^{at}\cdot\cos(\omega t)$	$\frac{s-a}{(s-a)^2+\omega^2}$
$e^{at}\cdot\sin(\omega t)$	$\frac{\omega}{(s-a)^2+\omega^2}$
$\sin(\omega t+\varphi)$	$\frac{s\sin\varphi+\omega\cos\varphi}{s^2+\omega^2}$
$\cos(\omega t+\varphi)$	$\frac{s\cos\varphi-\omega\sin\varphi}{s^2+\omega^2}$
$t^k\cdot e^{at}$	$\frac{k!}{(s-a)^{k+1}}$
$\sinh(\omega t)$	$\frac{\omega}{s^2-\omega^2}$
$\cosh(\omega t)$	$\frac{s}{s^2-\omega^2}$
$2\frac{\lvert z\rvert\cdot t^r}{r!}\cdot e^{-at}\cos(\omega t+\theta)$	$\frac{z}{(s+a-j\omega))^{r+1}}+\frac{\overline{z}}{(s+a+j\omega))^{r+1}}$
$2\frac{\lvert z\rvert\cdot t^r}{r!}\cdot e^{at}\cos(\omega t+\theta)$	$\frac{z}{(s-(a+j\omega))^{r+1}}+\frac{\overline{z}}{(s-(a-j\omega))^{r+1}}$

$f(t)$	$F(s)$
$t^n\cdot f(t)$	$(-1)^n\frac{\mathrm{d}^n}{\mathrm{d}s^n}F(s)$
$\delta(t)$	1
$\sinh(\omega t+\theta)$	$\frac{s\sinh(\theta)+\omega\cosh(\theta)}{s^2-\omega^2}$
$\cosh(\omega t+\theta)$	$\frac{s\cosh(\theta)+\omega\sinh(\theta)}{s^2-\omega^2}$
$(\alpha t+\beta)\cdot H(t+\lambda)$	$\frac{\beta e^{\lambda s}}{s}-\frac{\alpha e^{\lambda s}}{s^2}(s\lambda-1)$

Função do tempo	**Transformada**
$e^{at}\cdot f(t)$	$F(s-a)$
$f(t)$	$F(s)$
$f(at)$	$\frac{1}{a}\cdot F\left(\frac{s}{a}\right)$
$f(t-a)\cdot H(t-a)$	$e^{-as}\cdot F(s)$
$f^{(1)}(t)$	$sF(s)-f(0_-)$
$f^{(2)}(t)$	$s^2F(s)-sf(0_-)-f^{(1)}(0_-)$
$f(t)=f(t+T)$	$\frac{1}{1-e^{-sT}}\int_{0_-}^{T_-}e^{-st}\cdot f(t)\,\mathrm{d}t$
$\int_{0_-}^{t}f(\tau)\,\mathrm{d}\tau$	$\frac{F(s)}{s}$
$\int_{-\infty}^{t}f(\tau)\,\mathrm{d}\tau$	$\frac{F(s)}{s}+\frac{1}{s}\int_{-\infty}^{t}f(\tau)\,\mathrm{d}\tau$
$\frac{f(t)}{t}$	$\int_s^\infty F(\lambda)\,\mathrm{d}\lambda$

- Superamortecido:
  $\alpha^2>\omega_0^2$ ..... duas raízes reais e distintas

$$s_{1,2}=-\alpha\pm\sqrt{\alpha^2-\omega_0^2},\quad x(t)=K_1e^{s_1t}+K_2e^{s_2t}$$

- Amortecimento crítico:
  $\alpha^2=\omega_0^2$ ..... duas raízes reais e iguais

$$s_{1,2}=-\alpha,\quad x(t)=K_1e^{-\alpha t}+K_2te^{-\alpha t}$$

- Subamortecido (oscilatório):
  $\alpha^2<\omega_0^2$ ..... duas raízes complexas $(-\alpha\pm j\omega_d,\quad \omega_d=\sqrt{\omega_0^2-\alpha^2})$

$$s_{1,2}=-\alpha\pm j\sqrt{\omega_0^2-\alpha^2}=-\alpha\pm j\omega_d,\quad x(t)=K_1e^{-\alpha t}\cos(\omega_d t+\varphi)$$

$$Q=\frac{\omega_0}{2\alpha}$$

$$B=2\alpha$$

∗ RLC série

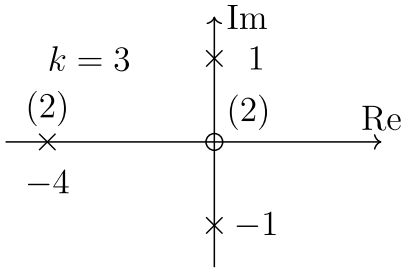
$$\alpha=\frac{R}{2L},\quad \omega_0^2=\frac{1}{LC}$$

∗ RLC paralelo

$$\alpha=\frac{1}{2RC},\quad \omega_0^2=\frac{1}{LC}$$

$$\ddot{x}(t) + 2\alpha\dot{x}(t) + \omega_0^2 x = f(t)$$

$$X = \underbrace{\frac{F(s)}{s^2 + 2\alpha s + \omega_0^2}}_{\text{resposta forçada}} + \underbrace{\frac{sx(0_-) + \dot{x}(0_-) + 2\alpha x(0_-)}{s^2 + 2\alpha s + \omega_0^2}}_{\text{resposta livre}}$$



$$G(s) = \frac{3s^2}{(s^2+1)(s+4)^2}$$

Multiplicidade (m) das raízes em s são indicadas por (m).

- $s = 0$ multiplicidade 2;
- $s = -4$ multiplicidade 2;
- $s = i$ multiplicidade 1;
- $s = -i$ multiplicidade 1;

$$\sin(2x) = 2\sin(x)\cos(x)$$

$$\cos(2x) = \cos^2(x) - \sin^2(x)$$

$$\tanh(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$$

$$\sinh(x) = \frac{e^{x} - e^{-x}}{2}$$

$$\sin(\omega t) = \frac{e^{j\omega t} - e^{-j\omega t}}{2}$$

$$\cosh(x) = \frac{e^{x} + e^{-x}}{2}$$

$$\cos(\omega t) = \frac{e^{j\omega t} + e^{-j\omega t}}{2}$$

$$\sin(p) + \sin(q) = 2\sin\left(\frac{p+q}{2}\right)\cos\left(\frac{p-q}{2}\right)$$

$$\cos(p) + \cos(q) = 2\cos\left(\frac{p+q}{2}\right)\cos\left(\frac{p-q}{2}\right)$$

$$\cos(p) - \cos(q) = -2\sin\left(\frac{p+q}{2}\right)\sin\left(\frac{p-q}{2}\right)$$

$$\sin(p) - \sin(q) = 2\sin\left(\frac{p-q}{2}\right)\cos\left(\frac{p+q}{2}\right)$$

$$|\sin(x) \pm \cos(x)| = \sqrt{1 \pm \sin(2x)}$$

Além disso, α é **denominado fator de amortecimento**, ω_0, **frequência própria não amortecida** e ω_d, **frequência própria amortecida**.